Exp 5a – Empirical Formula

INTRODUCTION

In this experiment, you will react solid magnesium with oxygen from the air to form a solid compound of magnesium and oxygen, an oxide of magnesium. Oxide compounds are a class of binary compounds of oxygen combined with another element. The goal is to experimentally determine the empirical formula (simplest whole-number ratio) of the oxide of magnesium. The unbalanced chemical equation (eqn 1) below represents the reaction of magnesium with oxygen. Note that in the formula of the oxide of magnesium x and y are used for the subscripts because they are considered unknown until they are determined using the results of this experiment.

$$\mathrm{Mg(s)} + \mathrm{O_2(g)} \rightarrow \mathrm{Mg}_x\mathrm{O}_y(s) \qquad (1)$$

BACKGROUND

The atom is the simplest form of an element. Compounds consist of atoms in specific ratios (water, H_2O). In fact, it is the ratio of atoms in a compound that yields its identity. This ratio of element atoms is responsible for the properties of the substance. Compounds composed of the same elements can have very different chemical and physical properties. A good example of this is hydrogen peroxide (H_2O_2) and water (H_2O). In its most concentrated form hydrogen peroxide will cause flesh to burst into flame on contact. In dilute solution, it is a disinfectant and a bleaching agent. Water does not possess these properties. Hydrogen peroxide has one more oxygen atom per molecule than water does. This small difference in the molecular formula results in large differences in chemical properties.

Compounds are generally split into two different categories: ionic and molecular. Molecular compounds are individual units with a specific number and arrangement of each type of atom, while ionic compounds are large continuous networks of atoms of the different types of elements. The number of atoms in a sample of an ionic compound varies depending on the size of crystal. An example of each type is shown below.

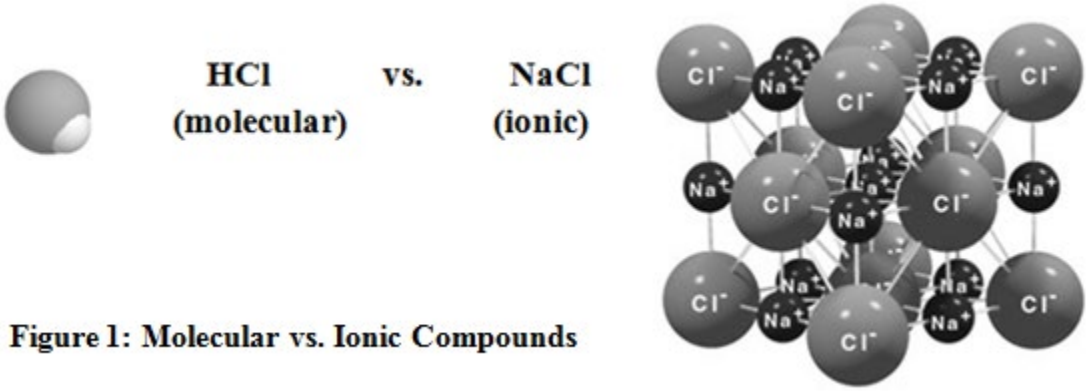


Figure 1: Molecular vs. Ionic Compounds

Since ionic compounds like NaCl (above) or Mg_xO_y have extremely large numbers of atoms that vary in quantity, the chemical formula is the smallest, simplest formula that repeats over and over. This simplest ratio is called the *empirical formula*. For molecular compounds, we write the *molecular formula*, which shows the actual number of atoms in each molecule. The molecular formula of a molecular compound is a multiple of the empirical formula of the molecular compound: the molecular formula of hydrogen peroxide H_2O_2 is twice its empirical formula HO, the molecular formula of water H_2O is once its empirical formula H_2O.

Experimental Error

As with all real-world experimentation there is a possibility that it will not work out exactly as we expect. The first step to eliminating errors is recognizing which errors are likely to occur. This experiment has a few well-known errors that may occur.

The first is incomplete combustion. It is possible that only the magnesium on the outside of the magnesium pieces reacts with oxygen. If this happens, unreacted metal will be left in the middle of the chunk of magnesium, changing the ratio of magnesium to oxygen in the final product. (Imagine if the NaCl crystal shown above had a few extra sodium atoms packed together in the middle that did not react with the chlorine gas.) To avoid this, break up the ash to expose unreacted magnesium and heat the sample over a strong flame.

A second common error that may occur is the reaction of magnesium with nitrogen instead of oxygen. The air in which the combustion of magnesium will take place is roughly 78% nitrogen and 21% oxygen. Consequently, even though the reaction with oxygen is strongly favored, it is likely that a small amount of magnesium will react with nitrogen according to the unbalanced equation (Eqn 2) shown below.

$$\mathrm{Mg(s)} + \mathrm{N(g)} \rightarrow \mathrm{Mg}_x\mathrm{N}_y(s) \qquad (5)$$

This reaction is impossible to avoid when heating in the open air and will likely represent a small amount of error in your final results.

* It is very important that scientists report only the results that were observed and do not color their results with what they "hoped" to observe. (Reporting altered or "faked" data has ended more than one scientist's career.) Even if you do not think of yourself as a scientist yet, remember that even if your results show significant errors, it is *always* better to report your actual results with an additional note about what errors you think may have occurred. It is never acceptable to try to make it seem like you observed something you did not.

ADDITIONAL READING

- Empirical and Molecular Formulas

SAFETY EQUIPMENT REQUIRED FOR LAB

- Goggles
- Lab coat
- Nitrile gloves (provided)

SUPPLY LIST

- Bench Post or Ring stand
- Iron ring
- Clay triangle
- Crucible
- Bunsen burner
- Wire mesh
- Steel wool
- Gas Hose
- Flint Lighter
- Crucible Tongs

CLASSROOM SUPPLY AND REACTANTS

- Magnesium shavings
- Scoopula

PROCEDURE

Cleaning the Crucible and Lid

1. ☐ Set up the apparatus as shown in Figure 2.
2. ☐ Obtain a porcelain crucible and lid.
3. ☐ Dump any solids out of the interior, scrape with steel wool, and wipe with a dry paper towel.

   ** Don't allow the crucible to come in contact with water.*
4. ☐ Place the crucible and lid on the stand. Leave the lid slightly ajar to allow air in. Heat over the flame until it glows slightly red-hot. Then set on the wire mesh to cool.
5. ☐ Allow the crucible to cool to room temperature (about 5 min)

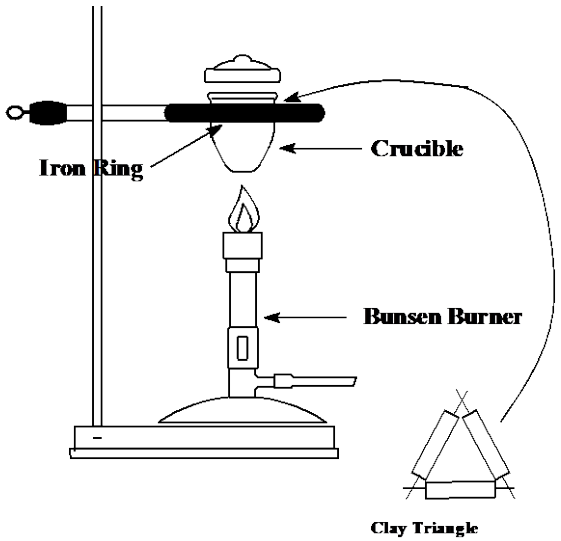


Figure 2: Crucible Heating

Data Collection

6. ☐ Record the mass of the empty crucible (Only the crucible! Not the lid! Remember to *always* record all digits shown on the balance.).

   ** Use the same balance to record all your mass measurements.*

   ** After you have obtained the first mass, if you must touch the crucible at any point, you must wear gloves to avoid leaving the oils from your fingers on the crucible and adding mass.*
7. ☐ On a piece of weighing paper, obtain between 0.10 to 0.15 g of magnesium turnings. This is just an approximate amount so there is no need to write down the value shown on the balance. Transfer the turnings to the crucible.
8. ☐ Tare (zero) the balance. Place the crucible back on the balance (without lid). Record the precise mass on your report sheet.
9. ☐ Heat the crucible with the lid slightly ajar to let air in to react with the magnesium. Use a hot flame to fully combust the magnesium for 5 minutes.

   ** If the magnesium should flare do not look directly at it. **
10. ☐ Remove the lid, continue to heat strongly for an additional 15 minutes. If the extra oxygen causes the magnesium to "flare up," place the lid back on it (ajar) temporarily.

11. ☐ After heating, set the crucible on a wire mesh and allow it to cool back to near room temperature. It takes about 5 minutes for the crucible to cool to room temperature.

    ** Do not set a hot crucible directly on the countertop or on paper. It will burn the counter or paper. Instead, when you remove it from the flame, set it on a wire mesh to allow it to cool. **

12. ☐ When the heating step is complete and the crucible has reached room temperature, obtain the mass of the crucible, and oxide of magnesium. (This is your 1st weighing.)

    * If you do not cool the crucible to room temperature, you will not be able to read a stable mass on the analytical balance. This is because the heat from the crucible causes air currents inside the balance.

13. ☐ Reheat the crucible as before, without the lid, for 10 minutes with the lid off and cool back to room temperature (about 5 minutes).

14. ☐ Once cooled to room temperature, re-weigh. (2nd weighing).

15. ☐ If your 1st and 2nd weights are within ±0.001 gram of each other, stop and do your calculations based on the higher of your two weights. If not, repeat step 13 until your last masses agree to within ±0.001 gram. Completely combusted magnesium oxide should be a white powder. Because of slight contamination, you should be looking for a light gray.

CLEANUP AND WASTE DISPOSAL

16. ☐ After checking your data with your instructor, empty the oxide compound into the “solids waste” container.

17. ☐ Scrape any excess from your crucible with steel wool before returning it to the supply bin.

DATA AND REPORT – DUE ____ / ____ / ____ **SECTION _____**

* Show your work and keep track of significant figures. **DATE _____**

PARTNER _____

DATA

Mass Values from Balance

1. Mass of Crucible (Empty) _____

2. Mass of Crucible + Mg _____

3. Mass of Crucible + MgxOy Compound _____ (1st weighing)

_____ (2nd weighing)

RESULTS

Calculated Masses

4. Mass of Mg used: _____

5. Mass of the MgxOy compound: _____

(use highest of your weighings)

6. Mass of Oxygen in MgxOy compound: _____

Empirical Formula

***** Carry extra significant figures on values that are used in subsequent calculations.

7. Calculate the moles of Mg initially present: _____

8. Calculate the moles of O present in the compound: _____

9. Report the mole ratio without reducing. Mg _____ O _____

   Example: Mg _*0.006243*_ *O* _*0.006151*_

10. Simplify the mole ratio from 9) as necessary and report the empirical formula of magnesium oxide:

    Example: Mg _*1.1*_ *O* _*1.0*_

_____

Percent Composition of Magnesium Oxide

$$\%mass = \frac{g\ element}{g\ compound} \times 100$$

11. Based on your experimental data (not your empirical formula), what is the percent Mg in your product?

_____

12. Based on your experimental data (not your empirical formula), what is the percent O in your product?

_____

POST-LAB QUESTIONS (USE VALUES FROM ONLINE REPORT)

1. Given the following analytical results: A _____ g sample of a compound was found to contain _____ g of _____ and _____ g of _____, determine the following.
   a. How many moles of each element are in the sample?

   b. What is the mass percent composition of each element in the compound, based on the experimental data provided above?

   c. What is the empirical formula for the compound?

   d. What is the correct name for the compound?

2. How can you tell from the chemical formula if a compound is expected to be an ionic compound or molecular compound?

3. What are structural differences between ionic compounds and molecular compounds?